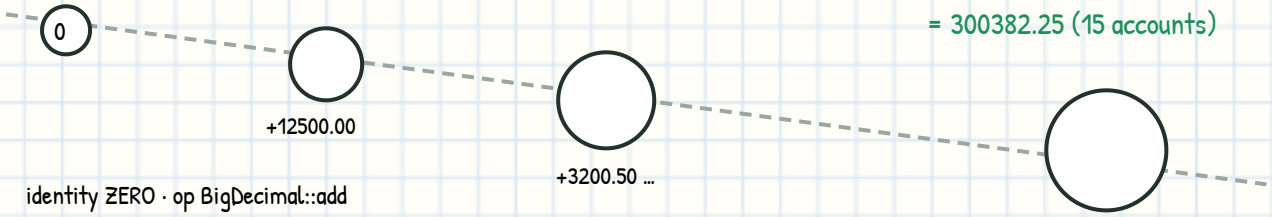
Terminal ops · my notes L09

the GO button decides what you get back

What kind of answer do I need? 🤔



reduce = a snowball rolling down the hill ⛄



Rule 1: associative

(a−b)−c ≠ a−(b−c)

`reduce(0, a − b)` : seq −55, parallel different 💥

Rule 2: true identity

x + id == x · id = 0, not 100

`reduce(100, sum)` : seq 155, parallel bigger 💥

3 forms

`reduce(id, op)` → T (empty: id) · `reduce(op)` → Optional (empty: empty) · `reduce(id, acc, combiner)` → U (Flight → seats 2078)

Exercise: month-end 📅 17

SGD, USD × 1.3456, rounded per line
Aug debits 4031.61 · Sep 7328.44 · Oct 5832.65
largest Oct debit: T042 1432.93

Numbers? simpler

`mapToInt(Flight::seats).sum()` → 2078

beats `reduce(0, acc, combiner)` for plain ints

Money 💰 and the empty box 📦

Average premium

sum 10826.50 ÷ 10

`divide(n, 2, HALF_UP)` → 1082.65

no scale → ArithmeticException on 1/3

Where do I round?

USD → SGD at 1.3456

per line: **1947.75** ✓ (exercise rule)

at the end: 1947.74

Empty stream = an empty classroom 🏫

(nobody here)

`List.of().stream()`

"Is ANYONE late?" → false

"Is EVERYONE on time?" → true (vacuous)

"Is NOBODY absent-minded?" → true

validation tip: "all payments approved" is **true** for an empty batch → also check `isEmpty()`

toList() 🔒

Java 16 · read-only · add() → UnsupportedOperationException · nulls OK

forEachOrdered 🚶

keeps order even in parallel: SQ318, SQ12, TR808 …

max → Optional

highest: T049 7721.99

lowest: T046 18.35

Dashboard, one terminal op per question

total balance 300382.25 (reduce) · highest T049 (max) · any overdrawn true (anyMatch) · avg premium 1082.65 (reduce + divide)

★ Remember

pick the op by the answer · toList read-only · reduce: associative + neutral identity · money: ZERO + add, divide with scale · empty: allMatch true · next → L10 Collectors I